

1. Circle the integral techniques that are most helpful for solving each of the following integrals and fill in the blank for the integral techniques you choose. You do not need to evaluate the integrals.

1). $\int \frac{\sin x}{\cos^2 x} dx$

☒ A. u -substitution with $u = \cos x$.

B. trigonometric substitution with $x =$

C. integration by parts with $u =$ and $dv =$

2). $\int \sqrt{x} \ln x dx$

A. u -substitution with $u =$.

B. trigonometric substitution with $x =$

☒ C. integration by parts with $u = \ln x$ and $dv = \sqrt{x}$

3). $\int \frac{x^2}{\sqrt{9-x^2}} dx$

A. u -substitution with $u =$.

☒ B. trigonometric substitution with $x = 3 \sin \theta$

C. integration by parts with $u =$ and $dv =$

2. Use partial fractions to evaluate $\int \frac{3}{(x-1)(2x+1)} dx$.

$$\frac{3}{(x-1)(2x+1)} = \frac{A}{x-1} + \frac{B}{2x+1}$$

$$3 = A(2x+1) + B(x-1)$$

$$3 = 2Ax + A + Bx - B$$

$$3 = (2A+B)x + A-B$$

$$\Rightarrow \begin{array}{l} 2A+B=0 \\ A-B=3 \end{array} \Rightarrow \begin{array}{l} 2A=3 \Rightarrow A=1 \\ B=-2 \end{array}$$

$$\int \frac{3}{(x-1)(2x+1)} dx = \int \frac{1}{x-1} dx + \int \frac{-2}{2x+1} dx = \ln|x-1| - 2 \cdot \frac{1}{2} \ln|2x+1| + C$$

$$= \ln|x-1| - \ln|2x+1| + C$$

$$\frac{3}{(x-1)(2x+1)} = \frac{A}{x-1} + \frac{B}{2x+1}$$

$$3 = A(2x+1) + B(x-1)$$

$$3 = 2Ax + A + Bx - B$$

$$3 = (2A+B)x + A - B$$

$$\Rightarrow \begin{aligned} 2A+B &= 0 \\ A-B &= 3 \end{aligned}$$

$$\Rightarrow 3A=3 \Rightarrow A=1$$

$$B=-2$$

$$\int \frac{3}{(x-1)(2x+1)} dx = \int \frac{1}{x-1} dx + \int \frac{-2}{2x+1} dx = \ln|x-1| - 2 \cdot \frac{1}{2} \ln|2x+1| + C$$

$$= \ln|x-1| - \ln|2x+1| + C$$